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Date: 2026-01-12

Intro to Finite Element Method (CIV E – 665)
Assignment No. 1



1 Solution to Question 1

First, integrate with respect to x :

$$\frac{dT}{dx} = C_1 \quad (1.1)$$

Where C_1 is an integration constant representing the temperature gradient.

Then, integrate with respect to x again:

$$T(x) = C_1 x + C_2 \quad (1.2)$$

Where C_2 is a second integration constant.

Equation 1.2 is the general solution.

1.1 Case (i) Temperature specified at the two boundaries

Based on this case, there are two fixed temperatures at both ends of the wall. Therefore, we can get two boundary conditions:

1. At $x = 0$, $T(0) = 80$
2. At $x = 1$, $T(1) = 150$

First, apply the first boundary condition to Equation 1.2, we can get:

$$T(0) = C_1 \times 0 + C_2 \Rightarrow C_2 = 80$$

Then, apply the second boundary condition to Equation 1.2 using $C_2 = 80$,

$$\begin{aligned} T(1) &= C_1 \times 1 + 80 = 150 \\ \Rightarrow C_1 &= 70 \end{aligned}$$

So,

$$T(x) = 70x + 80 \quad (1.3)$$

1.2 Case (ii) Temperature specified at $x = 0$ and Gradient specified at $x = 1$

In this case, the right side is insulated, which means no heat flows through it. We have these two boundary conditions:

1. At $x = 0$, $T(0) = 80$
2. At $x = 1$, $\frac{dT}{dx} = 0$

First, same as case (i), apply the first boundary condition to Equation 1.2, we can get:

$$\begin{aligned}T(0) &= C_1 \times 0 + C_2 \\ \Rightarrow C_2 &= 80\end{aligned}$$

Then, applying the second boundary condition with Equation 1.1, we can get $C_1 = 0$.

Therefore, substituting the constants back into Equation 1.2, the final solution is:

$$T(x) = 80$$

Which means the entire wall eventually reaches a uniform temperature of 80 °C.

2 Solution to Question 2

Known by the question, the governing equation is below

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = b(x) \quad (2.1)$$

Since the body force is assumed to be zero, Equation 2.1 becomes

$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = 0 \quad (2.2)$$

First, integrate both sides with respect to x

$$EA(x) \frac{du}{dx} = C_1$$

Based on the internal axial force is defined as stress times area, or $P(x) = \sigma A$, since $\sigma = E\epsilon$ and $\epsilon = \frac{du}{dx}$, so we can get

$$P(x) = EA \frac{du}{dx}$$

Comparing C_1 and $P(x)$, and no body forces, we can infer $F = P(x) = C_1$, right now

$$EA(x) \frac{du}{dx} = F$$

Rearrange the equation to solve for the strain $\frac{du}{dx}$, and substitute the given area function $A(x) = A_0(1 + \frac{x}{L})$:

$$\begin{aligned} \frac{du}{dx} &= \frac{F}{EA(x)} \\ &= \frac{F}{EA_0(1 + \frac{x}{L})} \end{aligned}$$

Integrate both sides with respect to x to get $u(x)$

$$\begin{aligned} u(x) &= \int \frac{F}{EA_0(1 + \frac{x}{L})} dx \\ &= \frac{F}{EA_0} \int \frac{1}{1 + \frac{x}{L}} dx \end{aligned}$$

To evaluate the integral $\int \frac{1}{1 + \frac{x}{L}} dx$, recall that $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln(ax+b)$. Here, $a = \frac{1}{L}$.

$$\begin{aligned} u(x) &= \frac{F}{EA_0} \left[L \cdot \ln \left(1 + \frac{x}{L} \right) \right] + C_2 \\ &= \frac{FL}{EA_0} \ln \left(1 + \frac{x}{L} \right) + C_2 \end{aligned}$$

Finally, apply the boundary condition of the left end: at $x = 0, u(x) = 0$

$$\begin{aligned} 0 &= \frac{FL}{EA_0} \ln \left(1 + \frac{0}{L} \right) + C_2 \\ &= \frac{FL}{EA_0} \ln(1) + C_2 \\ &\Rightarrow C_2 = 0 \end{aligned}$$

So,

$$u(x) = \frac{FL}{EA_0} \ln \left(1 + \frac{x}{L} \right)$$

3 Solution to Question 3

3.1 Find the expression for $w(x)$

The load diagram shows that there is a trapezoidal distributed load on the beam, so the load function varies linearly with x . Assume $w(x) = ax + b$.

Based on the question, $w_1 = -50$ and $w_2 = -1000$

$$\begin{cases} -50 = a \times 0 + b \\ -1000 = a \times L + b \end{cases} \Rightarrow \begin{cases} a = \frac{-950}{L} \\ b = -50 \end{cases}$$

Therefore,

$$w(x) = \frac{-950}{L}x - 50 \quad (3.1)$$

3.2 Mathematical expressions of the boundary conditions

At the left fixed end ($x = 0$), the displacement is zero, and the slope is zero.

$$y(0) = 0 \quad (3.2)$$

$$\theta(0) = \left. \frac{dy}{dx} \right|_{x=0} = 0 \quad (3.3)$$

At the right free end ($x = L$), the beam is free to move, so it has no internal forces, which means the bending moment and shear force are both zero.

Bending Moment:

$$M(L) = EI \left. \frac{d^2y}{dx^2} \right|_{x=L} = 0 \quad (3.4)$$

Shear Force:

$$V(L) = EI \left. \frac{d^3y}{dx^3} \right|_{x=L} = 0 \quad (3.5)$$

3.3 Solve for $y(x)$

The parameters follow are given: $EI = 1 \times 10^6 \text{ Nm}^2$ and $L = 1 \text{ m}$

Substitute $L = 1$ into Equation 3.1 from part i:

$$w(x) = -950x - 50$$

Since $EI \frac{d^4y}{dx^4} = w(x)$, the $y(x)$ can be obtained after integrating this equation four times.

Applying known boundary conditions to find the constants of integration

Comparing $EI \frac{d^4y}{dx^4} = w(x)$ and $V(x) = EI \frac{d^3y}{dx^3}$, the relationship between $V(x)$ and $w(x)$ can be found:

$$\begin{aligned} \frac{dV}{dx} &= \frac{d}{dx} \left(EI \frac{d^3y}{dx^3} \right) \\ &= EI \frac{d^4y}{dx^4} = w(x) \end{aligned}$$

so,

$$\begin{aligned} V(x) &= \int (-950x - 50) dx \\ &= -475x^2 - 50x + C_1 \end{aligned}$$

Apply boundary condition Equation 3.5

$$\begin{aligned} V(L) &= -475(1)^2 - 50(1) + C_1 = 0 \\ \Rightarrow -525 + C_1 &= 0 \\ \Rightarrow C_1 &= 525 \end{aligned}$$

so,

$$V(x) = -475x^2 - 50x + 525$$

Through integrating $V(x)$, $M(x)$ can be obtained,

$$\begin{aligned} M(x) &= \int V(x) dx = \int (-475x^2 - 50x + 525) dx \\ &= \frac{-475}{3}x^3 - 25x^2 + 525x + C_2 \end{aligned}$$

Apply boundary condition Equation 3.4,

$$\begin{aligned} -\frac{475}{3} \times 1^3 - 25 \times 1^2 + 525 \times 1 + C_2 &= 0 \\ \Rightarrow C_2 &= -\frac{1025}{3} \end{aligned}$$

so,

$$M(x) = \frac{-475}{3}x^3 - 25x^2 + 525x - \frac{1025}{3}$$

Through integrating $\frac{M(x)}{EI}$, $\theta(x)$ can be obtained,

$$\begin{aligned} \theta(x) &= \frac{\int \left(-\frac{475}{3}x^3 - 25x^2 + 525x - \frac{1025}{3} \right) dx}{EI} \\ &= \frac{-\frac{475}{12}x^4 - \frac{25}{3}x^3 + \frac{525}{2}x^2 - \frac{1025}{3}x + C_3}{EI} \end{aligned}$$

Apply boundary condition Equation 3.3, $C_3 = 0$ can be obtained, so,

$$\theta(x) = \frac{-\frac{475}{12}x^4 - \frac{25}{3}x^3 + \frac{525}{2}x^2 - \frac{1025}{3}x}{1 \times 10^6}$$

Integrate $\theta(x)$ with respect to x ,

$$y(x) = \int \theta(x) dx = \frac{-\frac{475}{60}x^5 - \frac{25}{12}x^4 + \frac{525}{6}x^3 - \frac{1025}{6}x^2 + C_4}{EI}$$

Apply boundary condition $y(0) = 0$, infer $C_4 = 0$, so

$$y(x) = \frac{-\frac{475}{60}x^5 - \frac{25}{12}x^4 + \frac{525}{6}x^3 - \frac{1025}{6}x^2}{1 \times 10^6}$$

3.4 Plot $y(x)$

use the Matplotlib module of python, $y(x)$ is shown as Figure 1

```

import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-0,1,1000)
y = (-475/60*x**5 - 25/12*x**4 + 525/6*x**3 - 1025/6*x**2)/(1e6)
plt.figure(figsize=(10,6))
plt.plot(x, y, label='Deflection y(x)')
plt.title('Beam Deflection Curve')
plt.xlabel('Position along the beam (x)')
plt.ylabel('Deflection (y)')
plt.legend()
plt.grid(True)
plt.show()

```

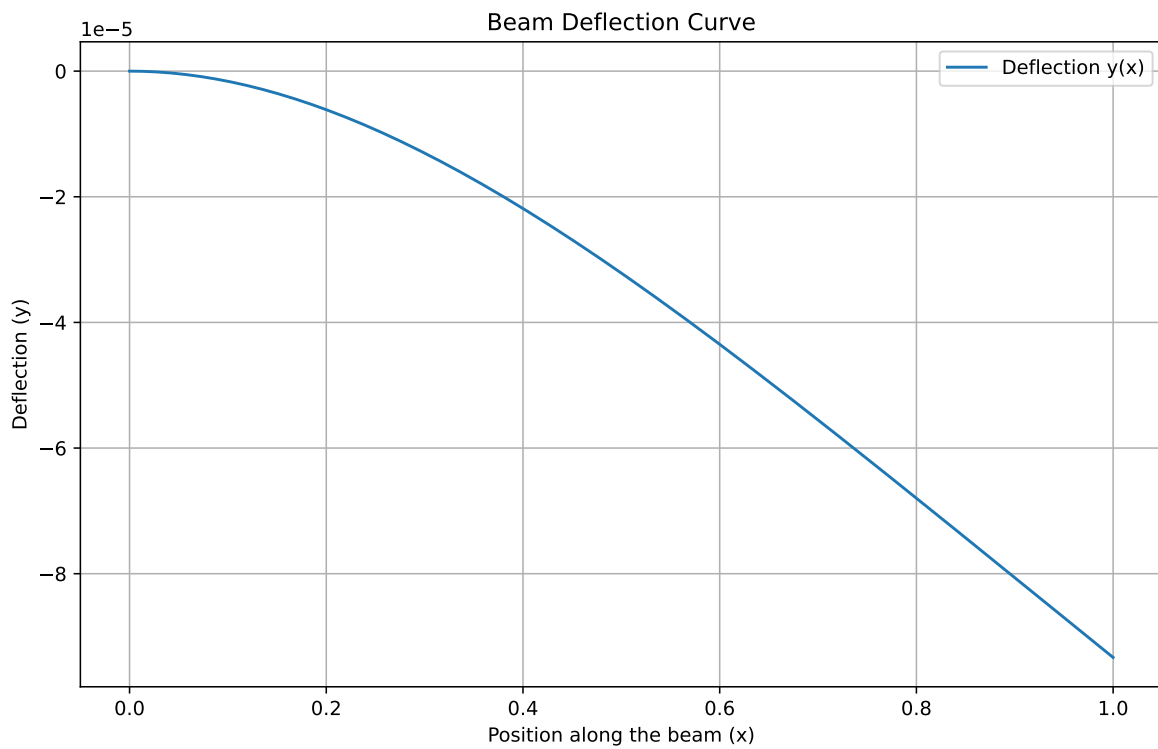


Figure 1: Deflection curve of the beam $y(x)$, in meters